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Project Title: Find 10 videos online showing objects moving from one place to another. Then, create a machine learning model that can accurately track the locations of these moving things in real-time.

Week 1 (21 September – 27 September)

The project was not assigned this week, so no work was done. We attended regular classes and waited for project allocation.

Week 2 (28 September – 4 October)

In the second week, the project was officially assigned, and our group was formed. After forming the team, I suggested that each member learn the basics of machine learning independently, so they could understand how data is processed, trained, and analyzed. We also decided to find 10 videos online that show moving objects, which would later be used for training and testing our model. I began practicing with basic Python libraries, such as NumPy and pandas, to become comfortable with data handling and analysis.

Week 3 (5 October – 11 October)

I practiced small projects using NumPy and pandas and watched tutorials on object detection. I found that **YOLOv8** and **OpenCV** could be used for real-time tracking. I built a small demo and shared it with my group members for learning purposes.

Week 4 (12 October – 18 October)

In the fourth week, I met with my instructor to discuss the project's progress. I explained that I used YOLOv8 and OpenCV for detection and tracking. However, my instructor advised me to focus more on the **machine learning concepts** themselves, such as understanding how models are trained, validated, and evaluated, rather than relying entirely on high-level libraries. Following that discussion, our group decided to explore how we could utilize OpenCV and machine learning more effectively in conjunction. We also talked about improving our understanding of model training, data annotation, and evaluation.

Week 5 (19 October – 25 October)

This week, our professor advised us not to use deep learning models for the project. Instead, we were encouraged to apply traditional machine learning methods to gain a deeper understanding of the core concepts. Based on this, we started looking into classical ML algorithms that could be used with OpenCV for object tracking.

Week 6 (26 October – 1 November)

I continued experimenting with tracking accuracy and improvements in detection. However, due to **midterm exams**, the project progress slowed as the group focused on studying.

Week 7 (2 November – Present)

After discussing with my group, we decided to learn **OpenCV**, **Kalman filters**, **SVM**, and **KNN** to strengthen our machine learning knowledge. Our goal is to develop our own tracking model, rather than relying fully on pre-trained tools. The **midterm exams are currently running**, so we are balancing exam preparation with continued project research and learning.